HydroDrone: A Circular External-Looking UAV-SAR Campaign for Short-Term Temporal Decorrelation Characterization

Pietro Grassi, *Member, IEEE*, Marco Manzoni, *Member, IEEE*, Stefano Moro, Andrea Monti-Guarnieri, *Senior Member, IEEE*, Julia Kubanek, and Stefano Tebaldini, *Senior Member, IEEE*

***Abstract*—This article presents the HydroDrone campaign, a dedicated uncrewed aerial vehicle (UAV)-based synthetic aperture radar (SAR) experiment designed to characterize short-term temporal decorrelation over vegetated scenes. The system employs a C-band frequency modulated continuous wave (FMCW) radar mounted on a multirotor UAV, performing repeated circular external-looking acquisitions over a chosen test site. This configuration enables the generation of dense interferometric stacks with temporal baselines ranging from tens of seconds to several hours. This work derives the specific mathematical framework for the computation of along-track wavenumbers and azimuth resolution in circular external-looking geometries, highlighting the fundamental differences with respect to standard linear trajectories. To compensate for residual trajectory errors arising from navigation uncertainties, a multiaperture interferometric trajectory compensation algorithm is introduced and applied across the stack. Preliminary results from a single campaign day are presented, including interferometric phase and coherence stacks before and after trajectory compensation. Coherence matrices computed over three different land-cover types (cornfield, medium-height vegetation, and meadow) are analyzed as a preliminary assessment of the coherence behavior and temporal sampling capability of the HydroDrone system. Thermal and spatial decorrelation contributions are quantified to confirm that the observed coherence loss is primarily driven by temporal decorrelation, providing a robust experimental basis for future vegetation clutter decorrelation modeling.**

***Index Terms*—Circular external-looking synthetic aperture radar (SAR), interferometry, temporal decorrelation, uncrewed aerial vehicle (UAV)-SAR, vegetation clutter.**

I. Introduction

SYNTHETIC aperture radar (SAR) systems represent a powerful tool for continuous environmental monitoring, owing to their ability to operate independently of solar illumination and weather conditions. However, in vegetated scenarios, the radar signal is strongly affected by the temporal variability of the scene: changes in the geometric and dielectric properties of the vegetation between repeated acquisitions give rise to temporal decorrelation, which manifests as a loss of interferometric coherence in interferometric SAR (InSAR) observations. Characterizing this phenomenon is essential both to correctly interpret the radar signal and to extract reliable geophysical parameters of interest [1], [2]. This is especially relevant in the context of future geosynchronous SAR missions, where the long synthetic aperture integration times make the characterization of short-term temporal decorrelation a critical prerequisite for reliable image formation [3].

In this context, the temporal evolution of vegetation clutter becomes a key aspect to be characterized. Radar clutter has been extensively studied since the early developments of radar technology [4], [5], yet its temporal behavior over vegetated areas remains poorly understood at short timescales. Most existing experimental datasets provide observations either at very short integration times, typical of airborne and low Earth orbit (LEO) SAR systems, or at relatively long revisit intervals on the order of minutes to days, leaving the intermediate regime (between a few seconds and several minutes) poorly characterized. Several experimental campaigns have been conducted to investigate radar backscatter variability and temporal decorrelation over vegetated scenes. Among SAR-based systems, the European Space Agency (ESA) HydroSoil campaign [6] employed a C-band ground-based SAR mounted on a rail to acquire repeated observations over agricultural fields, while SARSimHT-NG [7] relied on the airborne F-SAR system operating in C- and L-bands. The TropiScat [8] and BorealScat [9] experiments instead adopted tower-mounted fixed radar systems operating in P- and L-bands (TropiScat) and P-, L-, and C-bands (BorealScat), but lacked azimuth resolution and wide-area spatial coverage. Despite their contributions, all these campaigns share a common limitation in temporal sampling, with revisit intervals typically on the order of several minutes, leaving the intermediate regime between a few seconds and several minutes largely unexplored in a high-resolution imaging context.

To address this gap, the HydroDrone project has been conceived as a dedicated uncrewed aerial vehicle (UAV)-based SAR experiment specifically designed to measure short-term temporal decorrelation phenomena. Pre-

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Pietro Grassi, Marco Manzoni, Andrea Monti-Guarnieri, and Stefano Tebaldini are with the Department of Electronics, Information and Bioengineering (DEIB), Politecnico di Milano, 20133 Milan, Italy (e-mail: pietro.grassi@polimi.it).

Stefano Moro is with Aresys S.r.l., 20055 Milan, Italy.

Julia Kubanek is with European Space Agency (ESA), ESTEC, 2200 AG Noordwijk, The Netherlands.

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vious UAV-SAR experiments have demonstrated bistatic, multibaseline interferometric, and distributed multistatic concepts relevant to future spaceborne SAR systems [10], [11], [12]; HydroDrone is instead the first ESA-funded scientific UAV-SAR campaign specifically dedicated to the characterization of short-term temporal decorrelation over vegetated scenes. The primary objective of the campaign is to experimentally characterize the temporal evolution of interferometric coherence over timescales ranging from approximately 19 s to several hours and to assess its dependence on vegetation type and environmental conditions. The system exploits the flexibility of a C-band radar mounted on a UAV platform to perform repeated circular external-looking SAR acquisitions over the same heterogeneous scene, including different vegetation types and stable man-made structures, with extremely short revisit times. The repeated passes are designed to follow the same planned trajectory, ideally resulting in identical viewing geometries between consecutive acquisitions and thus negligible spatial decorrelation. Within a single flight, consisting of 30 passes followed by battery replacement, interferometric temporal baselines range from approximately 19 s (the duration of a single pass) to approximately 9 min; combining the three flights performed throughout the day extends the available temporal baselines to several hours.

In practice, small deviations from the ideal path occur due to UAV dynamics and navigation uncertainties. The portion of these deviations not captured by the onboard navigation system introduces phase errors in the interferograms, which are compensated using a coherent InSAR calibration strategy presented in this work. Once the trajectory corrections are estimated, the residual true baselines are expected to be small, but their impact on spatial decorrelation cannot be determined a priori and is therefore statistically quantified to ensure a reliable isolation of the temporal decorrelation component.

To the best of the authors' knowledge, the adoption of a circular external-looking acquisition geometry for repeated-pass interferometric analysis has not been extensively addressed in the literature. In this configuration, the along-track direction is not fixed as in the linear case, but varies pixel by pixel following the tangent to the circular trajectory at each target position. For this reason, the along-track wavenumbers and the corresponding resolution are explicitly derived and used to design a focusing strategy that guarantees a uniform along-track resolution for all pixels in the focused SAR images.

This article presents the HydroDrone experimental campaign and demonstrates its capability to measure short-term temporal decorrelation over different land-cover types. The main contributions of this work are summarized as follows.

1) Derivation of along-track wavenumbers and resolution for circular external-looking SAR geometries, explicitly highlighting differences with standard linear trajectories.
2) Extension of the interferometric trajectory compensation framework of [13] to circular geometries, including the retrieval of the along-track displacement component.
3) Experimental validation through UAV-SAR circular external-looking acquisitions.



Fig. 1. Optical satellite image of the test site (Bing Maps). The red dot indicates the center of the circular flight trajectory.

4) Computation of coherence matrices over different land-cover types to monitor the coherence evolution.
5) Statistical estimation of thermal and spatial decorrelation contributions to isolate the temporal component.

The remainder of this article is organized as follows. Section II describes the HydroDrone experimental campaign, including the test site and acquisition geometry, UAV-SAR platform, and ancillary measurements. Section III introduces the SAR and InSAR processing chain, including the derivation of azimuth resolution for the circular geometry, the interferometric phase calibration model, and the coherence matrix computation. Section IV presents InSAR results, including the interferometric stack before and after trajectory compensation, coherence matrices over different land-cover types, and the spatial coherence assessment. Finally, Section V summarizes the main findings and outlines future developments of the experiment.

II. HydroDrone Campaign

A. Objectives

The UAV-SAR HydroDrone campaign has been designed to experimentally characterize the short-term evolution of temporal decorrelation over timescales ranging from a few seconds to several hours. In this framework, the UAV platform provides a flexible solution, performing repeated circular external-looking acquisitions to maximize the covered area while achieving an extremely short revisit time. A key objective is to quantify the contribution of vegetation dynamics to coherence loss in interferometric observations, providing a comprehensive dataset for studying temporal decorrelation across different vegetation and environmental conditions.

B. Test Site and Acquisition Geometry

The experimental campaign is conducted over a radio-controlled airfield near Milan, Italy (Fig. 1), characterized by open terrain and compliance with local flight regulations for altitudes up to 120 m. The area exhibits a heterogeneous

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composition of natural and artificial targets, including grass meadows, bushes, and seasonal crops such as corn, alongside buildings and paved surfaces that provide stable reference targets for trajectory calibration.

In this environment, the UAV follows a circular external-looking trajectory with a radius of 7.5 m at a constant altitude of 120 m. This geometry was selected to repeatedly observe a wide surrounding area while maintaining a very short revisit interval. Moreover, consecutive passes are designed to ideally reproduce the same viewing geometry, thereby minimizing spatial decorrelation and facilitating the isolation of temporal decorrelation. For the approximately 1-km^2 imaged area considered in this work, a comparable linear acquisition would require a trajectory of approximately 2 km, corresponding to about 13.3 min at 2.5 m/s, compared with approximately 19 s for one circular pass. For the purposes of this work, three trajectory definitions are introduced: the ideal trajectory, programmed in the UAV control unit; the nominal trajectory, reconstructed from onboard sensors (GNSS/inertial measurement unit (IMU)/RTK) and used for SAR focusing; and the true trajectory, i.e., the actual physical path followed by the platform, which may deviate from the nominal one due to unmeasured perturbations.

These definitions are introduced to clearly separate trajectory planning, navigation-based reconstruction, and true platform motion effects within the interferometric processing chain.

C. Campaign Design

The project encompasses eight campaigns distributed throughout the year to capture seasonal variations in vegetation and soil conditions. For each season, two campaigns are performed approximately 45 days apart. Each campaign day consists of three flights (morning, mid-day, and afternoon), where each flight ideally comprises 30 consecutive circular passes. This strategy generates a dense interferometric stack with short-term lags ranging from approximately 19 s to approximately 9 min within a single flight, while cross-combining passes from different flight sessions allows for the observation of long-term lags spanning several hours.

D. UAV-Based SAR System

The radar sensor is integrated into an AntX T-Drones MX860 octocopter, which supports a 10-kg payload with a 25-min endurance. The platform's navigation relies on an IMU, dual GNSS antennas, and real-time kinematic (RTK) corrections to ensure centimetric positioning accuracy. The SAR sensor is a C-band frequency modulated continuous wave (FMCW) radar developed by Aresys S.r.l., operating at 5.8 GHz with a 150-MHz bandwidth, achieving a range resolution of approximately 1 m. The antenna's 80° azimuth aperture ensures a wide illumination sector during circular flight. Key system parameters are summarized in Table I. Note that the reported tangential resolution refers to the far-range case and thus represents the coarsest achievable value; as discussed in Section III-A, this resolution is enforced for all targets regardless of range by operating on the along-track wavenumber support during focusing.

TABLE I
MAIN SYSTEM AND ACQUISITION PARAMETERS

Parameter	Symbol	Value
Carrier frequency	f_0	5.8 GHz
Bandwidth	B	150 MHz
Azimuth beamwidth	α_{ant}	80°
Transmit power	P_{tx}	22 dBm
Flying altitude	h	120 m
Trajectory radius	r_s	7.5 m
Platform velocity	v	∼2.5 m/s
Range resolution	ρ_r	∼1 m
Tangential resolution	ρ_t	∼2 m

E. Ancillary Measurements

To support SAR data interpretation, a ground-based weather station monitors temperature, humidity, pressure, wind, and soil moisture, providing contextual information to be correlated with the observed decorrelation patterns. Furthermore, high-resolution ground-level photographs and orthophotos generated from dedicated UAV optical flights are used to define land-cover masks and provide a visual reference for the seasonal evolution of the vegetation.

III. SAR AND INSAR PROCESSING

This section describes the processing chain adopted for the HydroDrone dataset. First, the SAR image formation procedure is introduced, with particular focus on the derivation of along-track wavenumbers and resolution for the circular external-looking geometry. Then, the multiaperture interferometric trajectory compensation algorithm is presented. Finally, the computation of the interferometric coherence matrices is described.

A. SAR Image Formation

The first step in the InSAR processing chain is the formation of SAR images. The focusing procedure relies on the nominal trajectory, which represents the best available estimate of the platform's motion. For UAV-SAR systems, image formation is typically performed using the time-domain back-projection (TDBP) algorithm, which inherently accommodates nonlinear trajectories [14]. In the HydroDrone case, the adoption of TDBP is motivated by the circular flight geometry, which is intrinsically nonrectilinear even in ideal conditions. To mitigate the high computational cost of time-domain integration, the TDBP processor is implemented on a GPU using a CUDA-accelerated framework, significantly increasing processing throughput for large datasets [15].

The spatial resolution of an SAR system along a given direction is determined by the spectral support of the corresponding spatial wavenumber. We introduce a general Cartesian reference system (Fig. 2) where the sensor and target are located at $\mathbf{S}$ and $\mathbf{P}$, respectively. The line-of-sight (LOS) vector $\mathbf{R}$ and its magnitude R are defined as

$$\mathbf{R} = \mathbf{P} - \mathbf{S}, \quad R = \|\mathbf{R}\|, \quad \mathbf{r} = \frac{\mathbf{R}}{R}. \tag{1}$$

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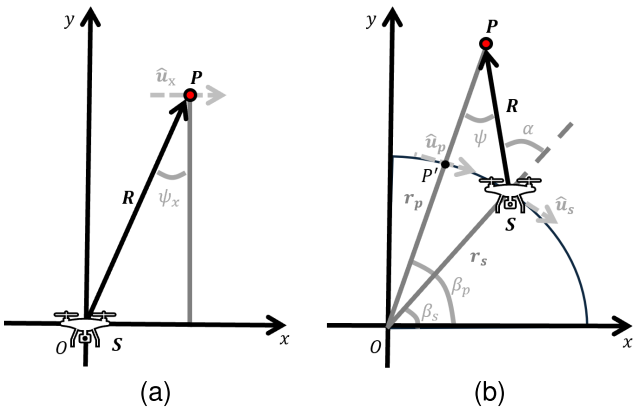


Fig. 2. Acquisition geometry and squint angle definition for (a) linear geometry (shown as reference) and (b) circular external-looking geometry adopted in the HydroDrone campaign. In both cases, the squint angle ψ is defined as the angle between the LOS vector $\mathbf{R}$ and the vector normal to the direction along which the resolution is evaluated ($\mathbf{u}_x$ in the linear case, $\mathbf{u}_p$ in the circular case).

Let $\mathbf{u}_p$ be a unit vector on the ground plane defining the direction along which the resolution is evaluated. The wavenumber K_u is the projection of the LOS unit vector onto $\mathbf{u}_p$

$$K_u = \frac{4\pi}{\lambda}\,\mathbf{r}\cdot\mathbf{u}_p. \tag{2}$$

The resolution ρ_u is then given by

$$\rho_u = \frac{2\pi}{\Delta K_u} \tag{3}$$

where ΔK_u is the variation of K_u over the synthetic aperture. While the range resolution is determined by the radar bandwidth, the along-track resolution can be controlled during focusing by properly selecting the wavenumber support for each pixel.

Enforcing a spatially uniform resolution is a critical prerequisite for the subsequent interferometric analysis: by ensuring that resolution cells have a consistent size across the entire scene, we maximize the statistical homogeneity of pixels in the coherence estimation, allowing for a reliable comparison of temporal decorrelation across different ranges and land-cover types. This general formulation is first applied to the standard linear trajectory as a reference and then extended to the circular geometry, as illustrated in Fig. 2.

1) Linear Geometry: For reference, the standard linear acquisition geometry is first considered as an illustrative example. In a standard linear acquisition aligned with the x-axis ($\mathbf{u}_x = [1, 0, 0]$), the along-track wavenumber is

$$K_x = \frac{4\pi}{\lambda}\frac{P_x - S_x}{R} = \frac{4\pi}{\lambda}\sin(\psi_x) \tag{4}$$

where λ is the wavelength, R is the slant range, P_x and S_x are the x-coordinates of the target and of the platform, respectively, and ψ_x is the instantaneous squint angle relative to the broadside direction for a fixed platform position [Fig. 2(a)]. Platform movement generates a variation in the squint angle $\Delta\psi_x$, which in turn generates a variation in the along-track wavenumber ΔK_x. From a sensitivity analysis, the along-track bandwidth is derived as

$$\frac{\Delta K_x}{\Delta\psi_x} = \frac{\partial K_x}{\partial\psi_x} \Rightarrow \Delta K_x = \frac{4\pi}{\lambda}\cos\psi_x\,\Delta\psi_x = \frac{4\pi}{\lambda}\frac{L_{SA}}{R}\cos\psi_x \tag{5}$$

where L_{SA} is the synthetic aperture length and R is the slant range of the target.

The along-track resolution ρ_x is then defined as

$$\rho_x = \frac{2\pi}{\Delta K_x} = \frac{\lambda R}{2L_{SA}\cos\psi_x}. \tag{6}$$

For a desired resolution ρ_x, and for small squint angles ψ_x, the synthetic aperture length L_{SA} processed during TDBP must be adjusted according to the range R of each pixel to guarantee uniform resolution across the scene

$$L_{SA} \approx \frac{\lambda R}{2\rho_x}. \tag{7}$$

The full antenna beamwidth α_{ant} provides the maximum achievable resolution

$$\rho_{x,\text{antenna}} = \frac{\lambda}{4\sin(\alpha_{ant}/2)}. \tag{8}$$

2) Circular External-Looking Geometry: In the circular external-looking configuration, the platform follows a circular trajectory of radius r_s centered at the origin O, as illustrated in Fig. 2(b). Let $\mathbf{u}_p$ denote the unit tangent vector to the trajectory at the target's angular position, defining the along-track (tangential) resolution direction.

The squint angle ψ is defined as the projection of the LOS unit vector onto the tangent

$$\sin(\psi) = \mathbf{r}\cdot\mathbf{u}_p \quad \Rightarrow \quad K_t = \frac{4\pi}{\lambda}\sin(\psi). \tag{9}$$

To relate the squint angle to the platform geometry, we introduce the illumination angle α between the LOS and the tangent at the platform's current position $\mathbf{u}_s$

$$\sin(\alpha) = \mathbf{r}\cdot\mathbf{u}_s. \tag{10}$$

Using the sine rule of the circular geometry, the relation between the two angles is

$$\sin(\psi) = \frac{r_s}{r_p}\sin(\alpha) \tag{11}$$

where r_p is the radial distance of the target from the trajectory center. Therefore, the tangential wavenumber can be expressed as

$$K_t = \frac{4\pi}{\lambda}\frac{r_s}{r_p}\sin(\alpha). \tag{12}$$

The tangential resolution ρ_t is defined as

$$\rho_t = \frac{2\pi}{\Delta K_t} \tag{13}$$

where ΔK_t is the wavenumber variation over the processed synthetic aperture.

For a desired resolution ρ_t, the synthetic aperture processed during TDBP must be adjusted for each pixel according to its radial distance r_p to guarantee uniform resolution across the scene. Note that, unlike the linear case, the tangential resolution in the circular geometry depends not only on the antenna beamwidth but also on the ratio between the target

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distance and the trajectory radius (r_p/r_s). This introduces a tradeoff between shorter revisit times, achieved with smaller radii, and finer resolution, obtained with larger radii.

Using the full antenna beamwidth α_{ant}, the reference tangential resolution is

$$\rho_{t,\text{antenna}} = \frac{\lambda}{4\sin(\alpha_{ant}/2)}\frac{r_p}{r_s}. \tag{14}$$

B. Trajectory Refinement via Multiaperture InSAR

Let the two passes forming the interferogram be referred to as primary and secondary. Any differential deviation of the true trajectories from the nominal paths used during focusing introduces phase errors in the interferograms. These errors are modeled following the framework introduced in [13], here extended to the full 3-D case to also retrieve the along-track displacement component, which was assumed known in the original formulation. Furthermore, details are reported in [16].

A key assumption for the validity of this model is that the SAR images are focused over sufficiently small subapertures. In this condition, for every pixel (x, y, z), the synthetic phase center can be considered stationary and coincident with the geometric center of the subaperture. Under this approximation, the interferometric phase error ϕ_{err} can be expressed as a linear combination of the 3-D residual displacements (dS_x, dS_y, dS_z) of the subaperture center

$$\phi_{err} = K_x dS_x + K_y dS_y + K_z dS_z \tag{15}$$

where the projection of the displacements onto the interferometric phase depends on the local imaging geometry through the wavenumbers $K_{x,y,z}$. While the displacement residuals (dS_x, dS_y, dS_z) are constant for a given subaperture, the wavenumbers vary pixelwise across the scene. In general, the model also includes a topographic contribution depending on the perpendicular baseline and the local terrain height; however, this term is negligible in the present experiment due to the flat nature of the test site, the small residual vertical baselines, and the low sensitivity to topography at C-band (as motivated in Section IV-C), and is therefore omitted.

The correction of the entire secondary trajectory is performed by dividing the nominal primary and secondary paths into multiple short overlapping subapertures, computing the phase center corrections for each, and then interpolating them to obtain a continuous correction along the complete trajectory. For the HydroDrone dataset analyzed in this work, subapertures with an along-track length of 1 m and an overlap of 88% were adopted, resulting in approximately 390 subapertures per circular pass.

In the case of multiple passes, the compensated secondary pass becomes the primary for the next interferometric pair, and the procedure is repeated. During a single campaign day, three acquisition flights consisting of 30 repeated passes each are performed. To calibrate all flights consistently, a common primary pass is selected for the three flights—namely the first pass of the first experiment.

For the circular external-looking geometry, it is convenient to define a local coordinate frame (t, r, z), where t is tangential (along-track), r is radial (from the trajectory center), and z is vertical. Given the platform angular position β_s along the circular path [Fig. 2(b)], the global wavenumbers K_x and K_y can be mapped onto local wavenumbers (K_t, K_r) via the rotation

$$\begin{bmatrix} K_t \\ K_r \end{bmatrix} = \begin{bmatrix} -\sin(\beta_s) & \cos(\beta_s) \\ \cos(\beta_s) & \sin(\beta_s) \end{bmatrix} \begin{bmatrix} K_x \\ K_y \end{bmatrix}. \tag{16}$$

The phase error can then be expressed in the local frame as

$$\phi_{err} = K_t dS_t + K_r dS_r + K_z dS_z. \tag{17}$$

1) Phase Center Correction Retrieval: The residual trajectory errors for an interferometric pair are denoted as

$$\Delta\mathbf{S} = \left[dS_t, dS_r, dS_z\right]^T. \tag{18}$$

The estimation of $\Delta\mathbf{S}$ is performed by evaluating a set of candidate corrections and selecting the one that best compensates the interferometric phase fringes. To this end, trial correction vectors are introduced as

$$\Delta\mathbf{S}_{trial} = \left[\delta_t, \delta_r, \delta_z\right]^T. \tag{19}$$

Let I_1 and I_2 denote the single-look complex (SLC) images of the primary and secondary passes, respectively, and define the interferogram as $\text{In} = I_1 I_2^*$.

The goal of the estimation is to compensate for the residual interferometric phase, ideally driving it to zero (because of neglected topographic contribution) over the processed subaperture interferogram.

For a given trial vector, the steered phase is computed using the local wavenumbers

$$\phi_{corr}(\Delta\mathbf{S}_{trial}) = K_t\,\delta_t + K_r\,\delta_r + K_z\,\delta_z \tag{20}$$

and its effect is evaluated by applying it to the corrupted interferogram.

A coarse estimate is obtained by maximizing the average power of the steered interferogram

$$\Delta\hat{\mathbf{S}}_c = \arg\max_{\Delta\mathbf{S}_{trial}} \left| \sum_k \text{In}(k)\ \exp\{-j\phi_{corr}(\Delta\mathbf{S}_{trial})\} \right|^2 \tag{21}$$

where the index k denotes the pixels within the considered subaperture interferogram. This cost function is unambiguous, since it is based on the squared magnitude, but it does not guarantee that the residual phase is exactly zero.

A fine estimate is then computed by maximizing the real part of the steered interferogram in a reduced search region around the coarse solution

$$\Delta\hat{\mathbf{S}}_f = \arg\max_{\Delta\mathbf{S}_{trial}} \sum_k \text{Re}\left\{\text{In}(k)\ \exp\{-j\phi_{corr}(\Delta\mathbf{S}_{trial})\}\right\}. \tag{22}$$

This cost function is maximized when the residual phase is zero, thus providing accurate phase compensation. However, it is inherently ambiguous due to its oscillatory behavior for large trial phase values. For this reason, the search is restricted to a small neighborhood around $\Delta\hat{\mathbf{S}}_c$, where the residual phase is already small, and the cost function exhibits a single well-defined maximum.

The final correction for the subaperture is obtained as

$$\Delta\hat{\mathbf{S}} = \Delta\hat{\mathbf{S}}_c + \Delta\hat{\mathbf{S}}_f. \tag{23}$$

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Corrections computed in the local polar frame (t, r, z) are transformed back to the global Cartesian frame. For each subaperture, the coarse-to-fine correction retrieval is iteratively repeated by updating the trajectory estimate until the magnitude of the newly estimated correction is smaller than 1 mm. For the HydroDrone dataset analyzed in this work, convergence was typically reached within ten iterations. The estimated corrections are then interpolated along the complete trajectory, and the SAR image is finally refocused using the updated trajectory.

C. Coherence Matrix Computation

The final stage of the processing chain is the generation of the interferometric coherence matrices, which serve as the primary tool for characterizing the temporal decorrelation across different land-cover types. After trajectory compensation and image refocusing, the coherence $\gamma_{i,j}$ between any two passes i and j in the stack is estimated directly.

In particular, for each region of interest (ROI) defined by selected field masks, the complex coherence is computed by averaging over the set $\mathcal{M}$ of L pixels

$$\gamma_{i,j} = \frac{\sum_{k\in\mathcal{M}} I_i(k)\, I_j^*(k)}{\sqrt{\sum_{k\in\mathcal{M}} |I_i(k)|^2 \sum_{k\in\mathcal{M}} |I_j(k)|^2}} \tag{24}$$

where $I_i(k)$ and $I_j(k)$ are the complex values of the kth pixel in the ith and jth compensated SLC images, respectively.

The resulting $P \times P$ (with P being the number of repeated InSAR passes) coherence matrix provides a complete temporal history of the scene.

1) Diagonal elements ($i = j$) are by definition equal to unity, representing the self-coherence of each acquisition.
2) Off-diagonal elements ($i \neq j$) capture the coherence decay as the temporal baseline increases from approximately 19 s up to approximately 9 min for a single flight and up to several hours when considering all flights performed during the day.

Since the focusing procedure enforces a spatially uniform along-track resolution across the scene, as described in Section III-A, all pixels contribute equally to the coherence estimation regardless of their distance from the trajectory center. By exploiting the high spatial resolution of the UAV-SAR platform, each field provides a large number of independent looks, which significantly reduces the estimation bias and provides a robust experimental foundation for the analysis presented in Section IV.

IV. Experimental Results: HydroDrone UAV-SAR Data

In this section, experimental results from the first campaign day in October 2025 are presented. The analysis is structured to demonstrate the effectiveness of the HydroDrone system and the consistency of the processing chain at different time scales. First, the InSAR stack for a single flight is presented, comparing results before and after residual trajectory calibration to highlight the impact of the proposed trajectory compensation algorithm. Subsequently, the consistency of the calibration across the entire day is evaluated through a multiflight stack, formed by a subset of two passes selected from each flight. Then, coherence matrices over different vegetated fields computed from all flights performed during the campaign day (Table II) are analyzed. Finally, a spatial coherence assessment is carried out to confirm that the observed coherence loss is primarily driven by temporal effects rather than by spatial decorrelation or SNR limitations. It is important to emphasize that this work focuses on demonstrating the system's capability to provide high-quality interferometric products, while a formal physical modeling and inversion of temporal decorrelation mechanisms are beyond the scope of this article and will be addressed in future studies.

TABLE II
Flight Schedule for the Campaign Day

Flight	Time of Day	Start Time
F001	Morning	10:39
F002	Mid-day	13:09
F003	Afternoon	14:10

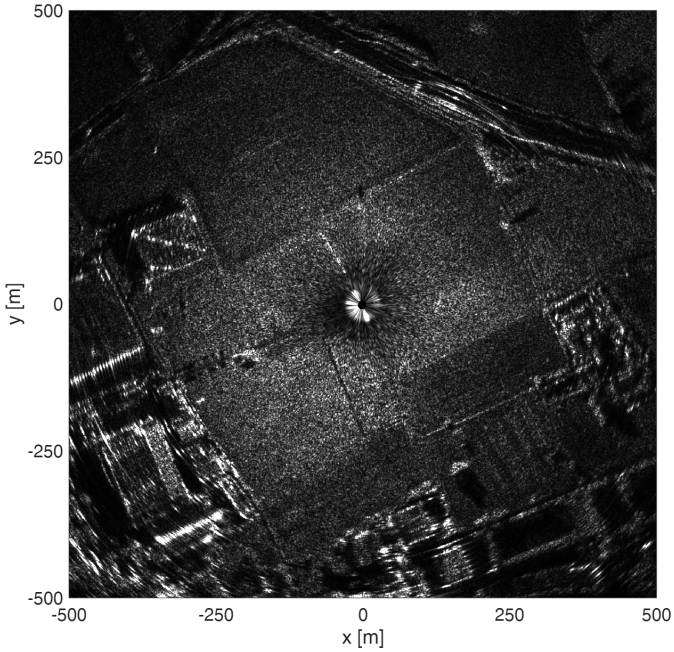


Fig. 3. SLC SAR image magnitude. Some residual blur is present due to navigation inaccuracies; however, the interferometric coherence is preserved as the focusing errors are similarly distributed across the image pair.

A. InSAR Stack and Calibration

The first step of the InSAR processing chain is the generation of the SLC images for the entire stack. Since the test site is predominantly flat (with the exception of a few buildings), all images are focused onto a constant-height grid.

Fig. 3 shows the magnitude of the SLC image for the first pass. Some residual blur, caused by inaccurate platform localization from the onboard sensors, is visible over bright targets in the form of prominent sidelobes. Nevertheless, as shown in the upper triangle of Fig. 4(a), the interferometric

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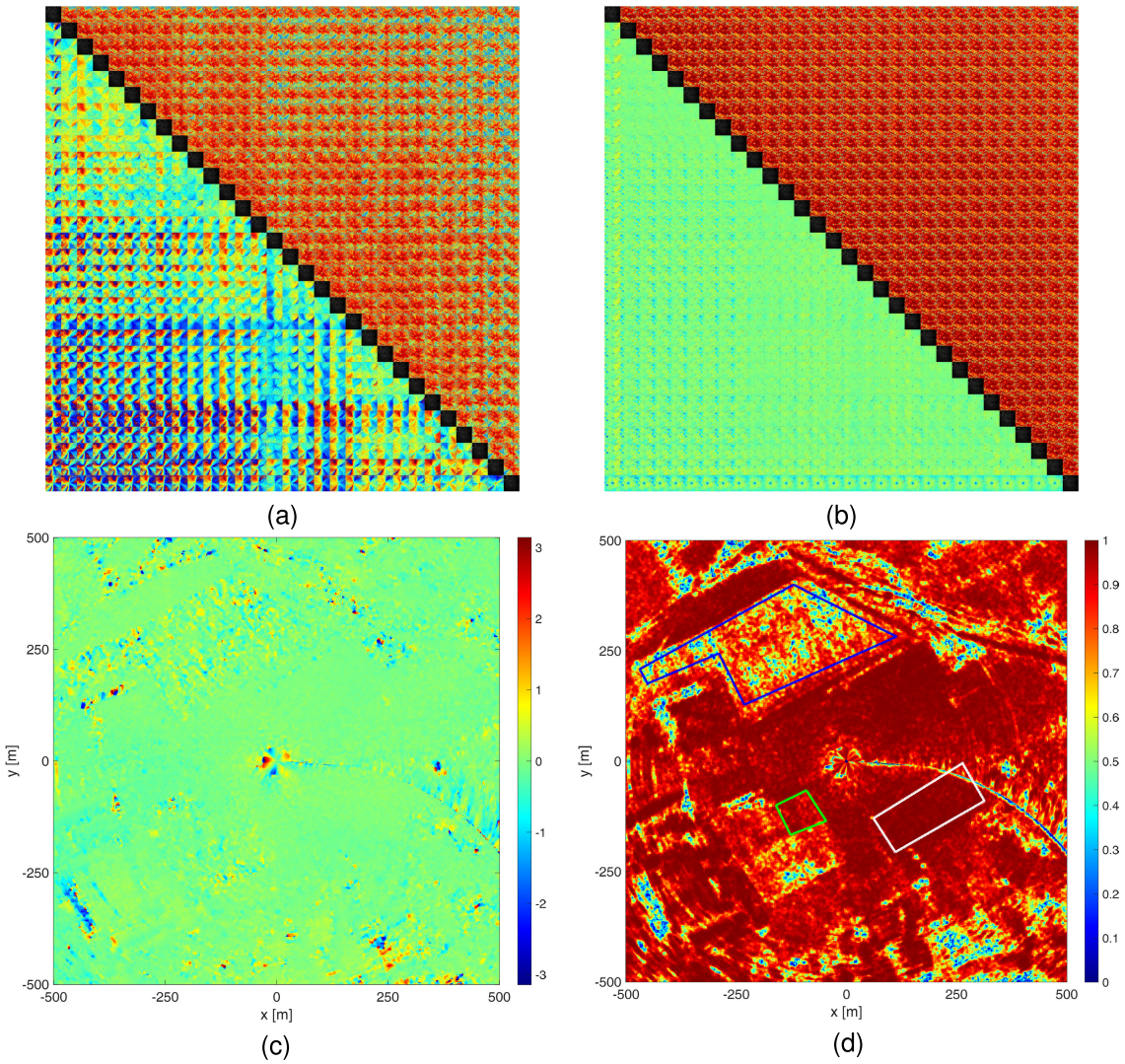


Fig. 4. (a) InSAR stack focused with nominal trajectories. (b) InSAR stack focused with calibrated trajectories. These matrices represent the interferometric phase (lower triangle) and coherence (upper triangle) for all combinations (30×30 interferograms). (c) and (d) InSAR phase and coherence for the first and second passes after calibration. Blue, green, and white boxes indicate the contours for the cornfield, medium-height vegetation, and meadow, respectively.

coherence is already sufficiently high to provide a reliable starting point for the coherent calibration. This is because the defocusing affects both SLC images in a pair in a similar manner; since the interferometric phase is a differential measurement, enough phase coherence is preserved to allow the multiaperture InSAR calibration algorithm to sense and correct the trajectory residuals, at least for the majority of the 30 repeated passes.

As expected, after the application of the trajectory refinement, the coherence values significantly increase across the entire stack, as shown in Fig. 4(b). This enhancement confirms the effectiveness of the calibration algorithm in compensating for residual unknown platform deviations. Fig. 4(c) and (d) provides an example of the interferometric phase and coherence computed from the first and second passes after calibration, showing a clean and flat phase over strong stable targets together with high coherence.

To further demonstrate the effectiveness of the calibration across the different flights, Fig. 5 reports a subset of the InSAR stack using two representative passes per flight. The time lag between adjacent passes within the same flight is on the order of 19 s, whereas the lag between different flights is several hours. This result demonstrates that the radar instrument maintains high phase stability across different flights. Such stability ensures that sufficient residual coherence is preserved even over long temporal baselines, allowing the InSAR calibration algorithm to effectively estimate and compensate for navigation drifts.

While stable targets maintain high coherence across flights, a clear decay is visible over vegetated areas as the temporal baseline increases. Some phase trends are present over certain targets, particularly in interferograms involving F002. As shown in Fig. 6, this is due to a small residual vertical baseline, which introduces sensitivity to uncompensated topography over taller man-made structures. However, as detailed in Section IV-B, this residual spatial decorrelation contribution is quantified based on the measured baseline and excluded to isolate the purely temporal decorrelation component.

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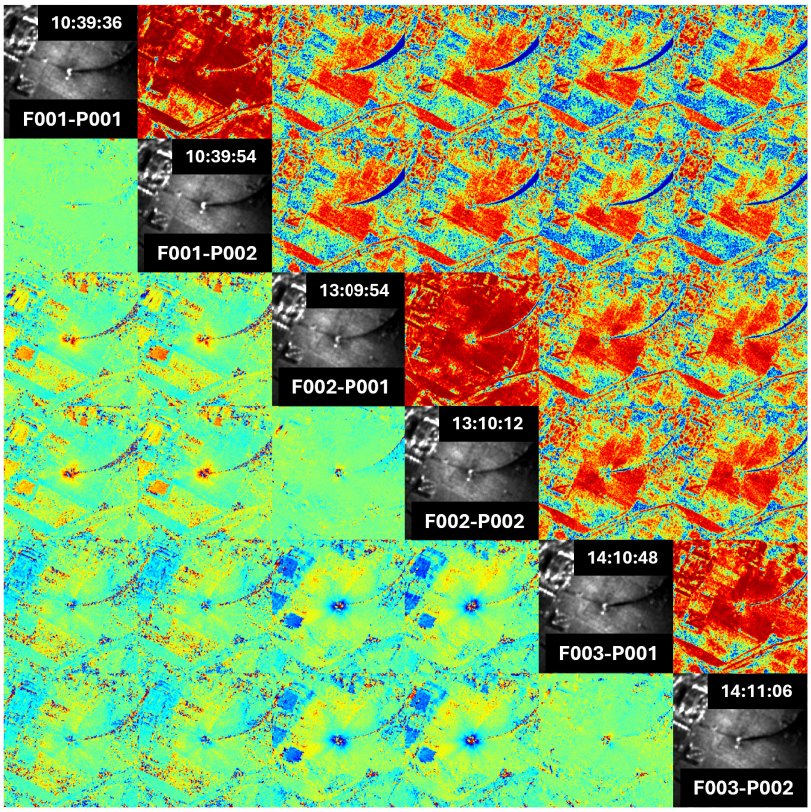


Fig. 5. InSAR stack for morning, mid-day, and afternoon flights. The main diagonal displays the SLC images, with the respective starting time, flight, and pass numbers labeled directly above each image. Temporally adjacent interferograms exhibit high coherence and nearly null phase, confirming the effectiveness of the trajectory compensation. Interferograms between different flights show significant coherence decay due to temporal decorrelation. Residual phase trends over buildings remain visible in interferograms between different flights, reflecting the topographic sensitivity induced by the not exactly zero perpendicular baseline.

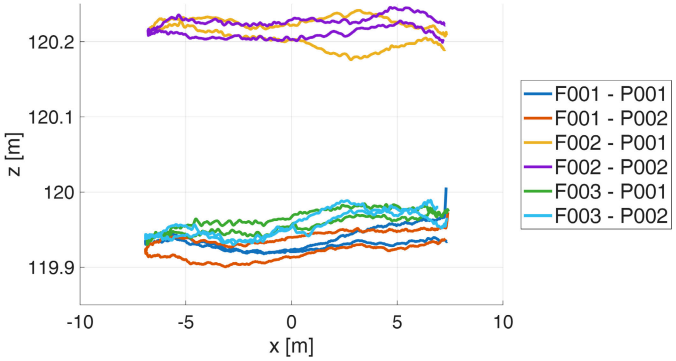


Fig. 6. Residual vertical baseline for the different passes (Pxxx) acquired during the three flights (Fxxx) of the presented campaign day.

B. Coherence Analysis Over Different Vegetated Fields

The analysis presented in this section is intended as a preliminary assessment of the coherence behavior observed during the first campaign day and as a validation of the temporal sampling capability of the HydroDrone system. The coherence matrices are therefore discussed qualitatively, whereas a quantitative characterization of the coherence decay, including model selection, estimation of decorrelation time constants, and physical interpretation of the retrieved parameters, is beyond the scope of the present work and will be addressed in a dedicated study.

TABLE III
VEGETATION HEIGHT OF SELECTED FIELDS

Vegetation Type	Height [m]
Corn	2.0
Medium-height vegetation	1.0
Meadow	0.0

Three distinct vegetated fields are considered [Fig. 4(d)]: a cornfield, medium-height vegetation, and a meadow. Fig. 7 displays the optical images of the selected fields obtained from the orthophoto and a qualitative map of vegetation heights through the digital surface model (DSM) acquired during the campaign. Note that the DSM exhibits a systematic offset of approximately 1.5 m with respect to the ground level. Table III summarizes the nominal vegetation heights for these fields.

The interferometric coherence matrices, computed as the mean value over the masked pixels, are shown in Fig. 8(a) for flight F001 and in Fig. 8(b) for the full daily set (F001–F003). Note that a few passes were excluded from the analysis, as they were corrupted by significant navigation errors that prevented reliable trajectory refinement.

The matrices reveal that coherence drops in the cornfield are severe even at short time lags, with an almost complete coherence loss after a few hours. On the contrary, the meadow and medium-height vegetation exhibit much higher stability,

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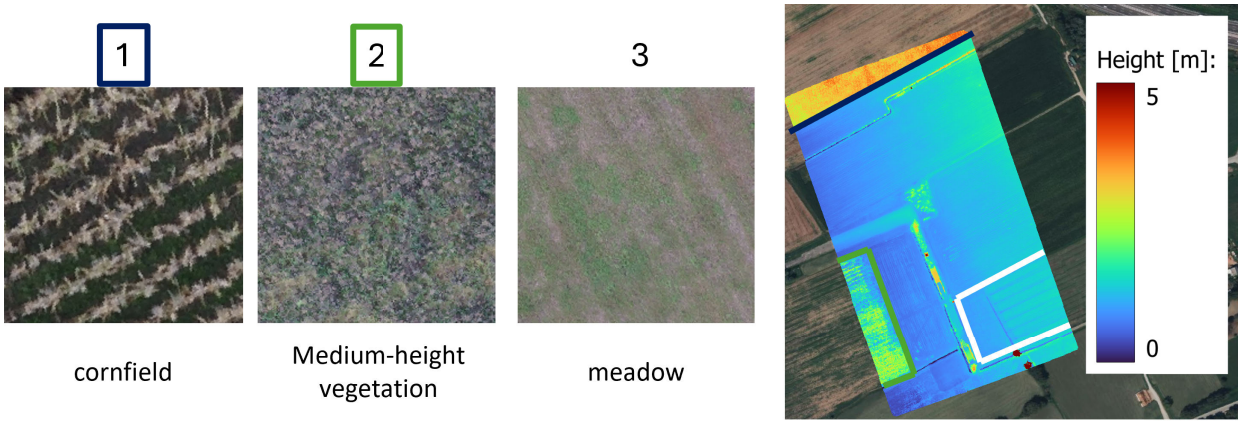


Fig. 7. Selected fields for coherence analysis (highlighted with blue, green, and white boxes). The right panels show the DSM and orthophoto acquired as ancillary data during the campaign, used to characterize the vegetation types.

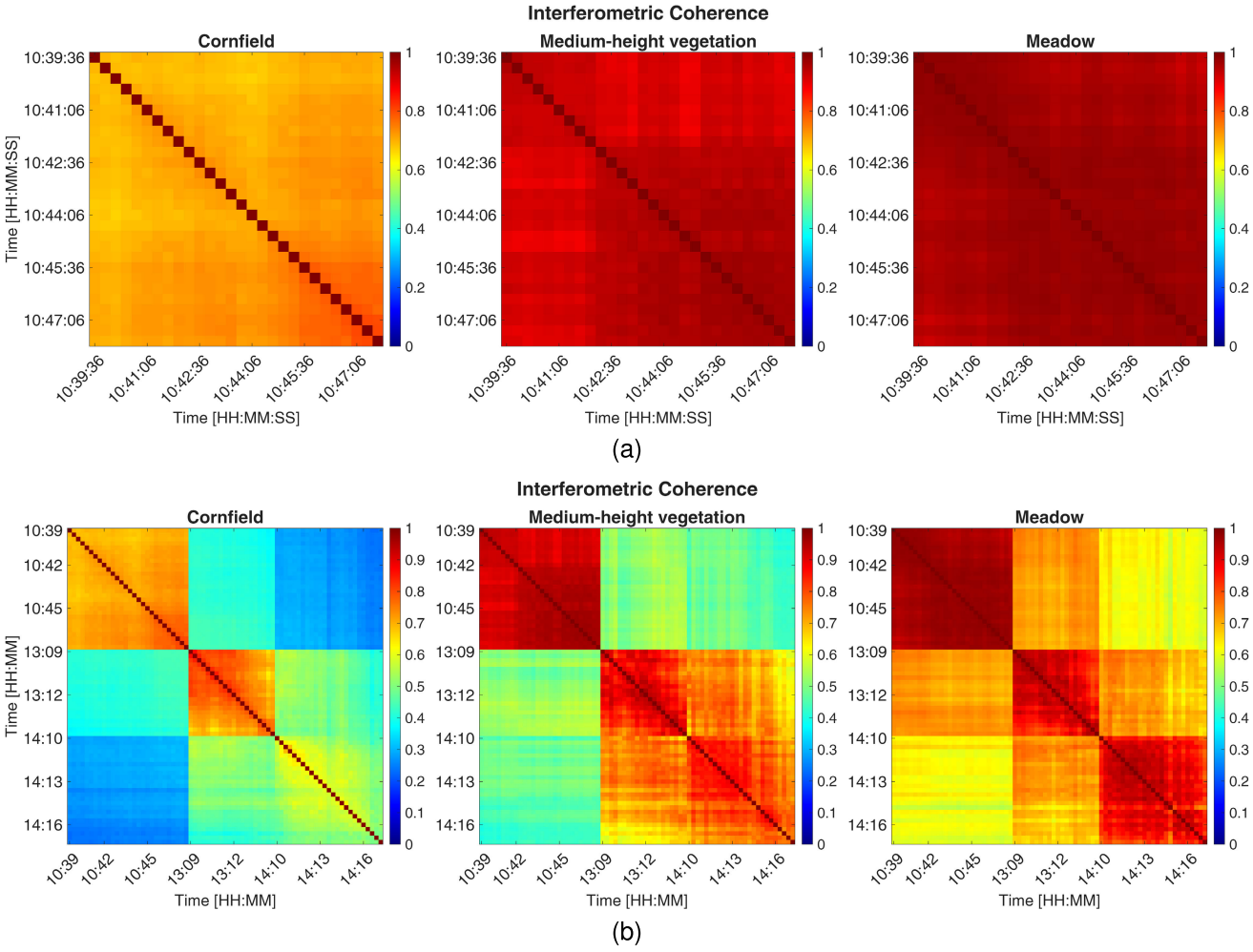


Fig. 8. Interferometric coherence matrices for the three selected fields. (a) Single flight F001 and (b) full daily dataset (F001, F002, and F003).

maintaining high coherence levels within a single flight and showing a more gradual decay over the day. In particular, for the cornfield and medium-height vegetation, the coherence between F002 and F003 is higher than that between F001 and F002, consistent with the shorter temporal separation (~1 h compared to ~2.5 h). These observations are consistent with the expected behavior of vegetation scatterers at C-band, and are reported here as a qualitative validation of the measurement chain rather than as a quantitative physical analysis.

A key aspect of the dataset is the diversity of temporal baselines it provides, spanning from approximately 19 s between consecutive passes within a single flight up to several hours across the full daily acquisition. This range allows the system to capture both the rapid decorrelation dynamics of

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fast-moving vegetation, such as the cornfield, and the slower evolution of more stable targets, demonstrating the ability of the HydroDrone platform to densely sample the coherence temporal profile across heterogeneous land-cover types. These results therefore provide a robust experimental foundation for future temporal decorrelation modeling across a broad range of timescales and vegetation conditions.

C. Thermal and Spatial Coherence Assessment

To ensure that the observed coherence loss is dominated by temporal effects, all other sources of coherence loss must be quantified. The total interferometric coherence magnitude can be expressed as the product of several contributing factors [17], [18]

$$\gamma = \gamma_{SNR}\,\gamma_{spatial}\,\gamma_{temp} \tag{25}$$

where $\gamma_{spatial}$ represents the combined geometric and volumetric coherence contribution, and γ_{temp} is the temporal coherence of interest.

The thermal noise contribution is characterized by

$$\gamma_{SNR} = \frac{1}{1+\mathrm{SNR}^{-1}}. \tag{26}$$

Based on the system parameters, the noise equivalent sigma zero (NESZ) is approximately −27 dB at a slant range of 500 m, ensuring an adequate signal-to-noise ratio for all target types present in the scene, including bare soil, grassland, and agricultural fields [19]. Since the worst case SNR is approximately 15 dB (corresponding to $\gamma_{SNR} \approx 0.97$), the thermal noise contribution is considered negligible and γ_{SNR} is set to unity in the following analysis.

The spatial coherence $\gamma_{spatial} = \gamma_{geom}\,\gamma_{vol}$ accounts for decorrelation arising from differences in the viewing geometry between repeated passes. The geometric component is parameterized by the perpendicular baseline $b_\perp$ as

$$\gamma_{geom} = 1 - \frac{b_\perp}{b_\perp^c} = 1 - \frac{f_0\,\Delta\theta}{B\tan(\theta)} \tag{27}$$

where f_0 is the carrier frequency, B is the signal bandwidth, θ is the incidence angle, and $\Delta\theta$ is the difference in incidence angle between the two passes. The critical baseline $b_\perp^c$ defines the threshold beyond which complete geometric decorrelation occurs.

The volumetric component accounts for the presence of a vertically distributed scattering volume. For a uniformly distributed layer of height h_v, it can be expressed as

$$\gamma_{vol} = \left|\mathrm{sinc}\left(\frac{\Delta k_z\,h_v}{2}\right)\right| \tag{28}$$

where $\mathrm{sinc}(x) = \sin(x)/x$, and Δk_z is the interferometric vertical wavenumber defined as

$$\Delta k_z = \frac{4\pi}{\lambda}\frac{b_\perp}{R\sin(\theta)} \tag{29}$$

with R denoting the slant range.

The uniform vertical scattering profile assumed in (28) represents a simplified approximation of the actual vegetation structure, particularly for the cornfield, whose vertical reflectivity distribution may be nonuniform. In the present analysis, the sinc model is not intended to provide a detailed physical description of the vegetation canopy, but only an indicative estimate of the volumetric decorrelation associated with the residual baseline variations of the UAV acquisitions. The resulting volumetric coherence values should therefore be interpreted as approximate estimates under the uniform-layer assumption.

Small perpendicular baselines are thus required to minimize spatial decorrelation, which is further mitigated at C-band due to the frequency dependence of Δk_z.

Although the repeat-pass strategy targets a zero-baseline configuration, platform deviations result in small, nonzero perpendicular baselines that vary for each pass. Based on previous flight experience, residual perpendicular baselines typically remain within 15 cm; however, larger deviations were observed in the presented campaign day, as shown in Fig. 6. For this reason, $\gamma_{spatial}$ is evaluated pixel-by-pixel for each interferometric pair using the actual $b_\perp$ estimated from the refined trajectories. The temporal coherence component can then be isolated as

$$\gamma_{temp} = \frac{\gamma}{\gamma_{SNR}\,\gamma_{spatial}}. \tag{30}$$

The resulting mean and standard deviation values ($\mu \pm \sigma$) of $\gamma_{spatial}$ are reported in Table IV for each flight and for the entire daily dataset. Within the adopted uniform-layer approximation, the high values (mostly > 0.95) and the extremely low standard deviations indicate that spatial decorrelation is not the dominant contribution, enabling a focused analysis of temporal decorrelation.

TABLE IV
MEAN SPATIAL COHERENCE AND STANDARD DEVIATION ($\mu \pm \sigma$) FOR THE SELECTED FIELDS

Exp. (Start Time)	Corn	Med. Veg.	Meadow
F001 (10:39)	0.9948 ±0.0054	0.9908 ±0.0082	0.9952 ±0.0037
F002 (13:09)	0.9570 ±0.0486	0.9535 ±0.0488	0.9882 ±0.0104
F003 (14:10)	0.9732 ±0.0297	0.9670 ±0.0307	0.9897 ±0.0075
Full daily dataset	0.9550 ±0.0543	0.9484 ±0.0548	0.9860 ±0.0106

V. CONCLUSION

This work presented the HydroDrone UAV-SAR campaign and the dedicated processing chain developed for SAR focusing and interferometric calibration.

A specific mathematical framework for the derivation of tangential wavenumbers and along-track resolution in circular external-looking geometries was introduced, highlighting the key differences with respect to conventional linear SAR trajectories. This formulation enables the focusing of SAR images with a spatially uniform along-track resolution, providing a consistent and robust basis for subsequent interferometric analysis. Residual trajectory errors arising from navigation

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uncertainties were effectively compensated using the proposed multiaperture InSAR calibration algorithm.

Experimental results from the campaign day in October 2025 demonstrate the validity of this approach. The consistency of the InSAR compensation was verified across multiple temporal scales, from intraflight acquisitions to interflight passes spanning several hours. The quality of the interferometric products is confirmed by the high coherence observed over stable targets and by the near-zero interferometric phase obtained after trajectory refinement, consistent with the near-zero-baseline configuration. A preliminary analysis of coherence matrices was conducted over three distinct vegetated fields (cornfield, medium-height vegetation, and meadow), providing an overview of coherence behavior across repeated passes.

The results demonstrate the capability of the HydroDrone system to capture decorrelation dynamics over time lags ranging from tens of seconds to several hours. Furthermore, the high spatial coherence observed across the dataset indicates that the measured coherence loss is primarily driven by temporal effects rather than by spatial decorrelation or SNR limitations.

We remark that the results presented in this article are to be interpreted as a validation of the experimental configuration and processing framework rather than as a complete characterization of temporal decorrelation phenomena. An exhaustive analysis of temporal dynamics across the eight campaign days will be addressed in future work.

The successful validation of the system during this first campaign day establishes a solid experimental foundation for future investigations. Building on the full set of eight planned campaigns, future work will focus on the detailed analysis and physical modeling of temporal decorrelation across different vegetation types. This will enable the characterization of environmental and agricultural dynamics, supported by extensive ancillary ground measurements.

REFERENCES

[1] W. Wagner et al., "Temporal stability of soil moisture and radar backscatter observed by the advanced synthetic aperture radar (ASAR)," *Sensors*, vol. 8, no. 2, pp. 1174–1197, Feb. 2008. [Online]. Available: https://www.mdpi.com/1424-8220/8/2/1174

[2] A. Monti-Guarnieri, M. Manzoni, D. Giudici, A. Recchia, and S. Tebaldini, "Vegetated target decorrelation in SAR and interferometry: Models, simulation, and performance evaluation," *Remote Sens.*, vol. 12, no. 16, p. 2545, Aug. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/16/2545

[3] M. Monti, A. M. Guarnieri, F. Pelliccia, and A. Renga, "Geostationary interferometric SAR: Orbit design and control implementation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5211017, doi: 10.1109/TGRS.2025.3565889.

[4] J. B. Billingsley, *Low-Angle Radar Land Clutter: Measurements and Empirical Models*. Stevenage, U.K.: William Andrew Publishing, 2002.

[5] P. C. Dubois, J. van Zyl, and T. Engman, "Measuring soil moisture with imaging radars," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 4, pp. 915–926, Jul. 1995. [Online]. Available: https://ieeexplore.ieee.org/document/406677/

[6] A. Aguasca et al., "Hydrosoil, soil moisture and vegetation parameters retrieval with a C-band GB-SAR: Campaign implementation and first results," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 6685–6687. [Online]. Available: https://ieeexplore.ieee.org/document/9554572/

[7] A. Balenzano et al., "SARSimHT-NG campaign over southern Italy for investigating sub-daily water processes," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 4376–4379. [Online]. Available: https://ieeexplore.ieee.org/document/10641490/

[8] C. Albinet et al., "TropiSCAT: A ground based polarimetric scatterometer experiment in tropical forests," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 3, pp. 1060–1066, Jun. 2012. [Online]. Available: https://ieeexplore.ieee.org/document/6216452/

[9] A. R. Monteith and L. M. H. Ulander, "Temporal survey of P- and L-band polarimetric backscatter in boreal forests," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3564–3577, Oct. 2018. [Online]. Available: https://ieeexplore.ieee.org/document/8335779/

[10] L. Carrer et al., "Distributed drone-based demonstrator for SAR: The first results," in *Proc. 16th Eur. Conf. Synth. Aperture Radar (EUSAR)*, Baden-Baden, Germany, 2026, pp. 47–51.

[11] S.-Y. Jeon et al., "Experimental demonstration of bistatic UAV-borne SAR and InSAR," in *Proc. EUSAR 15th Eur. Conf. Synth. Aperture Radar*, Apr. 2024, pp. 1150–1155. [Online]. Available: https://ieeexplore.ieee.org/document/10659473

[12] V. Mustieles-Perez, "Experimental demonstration of UAV-based ultra-wideband multi-baseline SAR interferometry," in *Proc. 15th Eur. Conf. Synth. Aperture Radar*, Apr. 2024, pp. 1156–1161. [Online]. Available: https://ieeexplore.ieee.org/document/10659455

[13] S. Tebaldini, F. Rocca, M. M. d'Alessandro, and L. Ferro-Famil, "Phase calibration of airborne tomographic SAR data via phase center double localization," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1775–1792, Mar. 2016. [Online]. Available: http://ieeexplore.ieee.org/document/7308057/

[14] O. Frey, C. Magnard, M. Ruegg, and E. Meier, "Focusing of airborne synthetic aperture radar data from highly nonlinear flight tracks," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1844–1858, Jun. 2009. [Online]. Available: http://ieeexplore.ieee.org/document/4812049/

[15] Y. Yu, M. M. d'Alessandro, S. Tebaldini, and M. Liao, "Signal processing options for high resolution SAR tomography of natural scenarios," *Remote Sens.*, vol. 12, no. 10, p. 1638, May 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/10/1638

[16] M. Manzoni, P. Grassi, L. Mantuano, and S. Tebaldini, "Phase calibration of UAV-SAR tomographic data with 3D phase center double localization," in *Proc. 16th Eur. Conf. Synth. Aperture Radar (EUSAR)*, Baden-Baden, Germany, 2026, pp. 1048–1053.

[17] R. Ahmed, P. Siqueira, S. Hensley, B. Chapman, and K. Bergen, "A survey of temporal decorrelation from spaceborne L-band repeat-pass InSAR," *Remote Sens. Environ.*, vol. 115, no. 11, pp. 2887–2896, Nov. 2011. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0034425711001349

[18] F. Gatelli, A. Monti Guamieri, F. Parizzi, P. Pasquali, C. Prati, and F. Rocca, "The wavenumber shift in SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 855–865, Jul. 1994. [Online]. Available: https://ieeexplore.ieee.org/document/298013/

[19] F. T. Ulaby and D. G. Long, *Microwave Radar and Radiometric Remote Sensing*. Ann Arbor, MI, USA: Univ. of Michigan Press, 2014.

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